

Riccardo Orlando

PH.D. IN DEEP LEARNING AND NATURAL LANGUAGE PROCESSING

☎ (+39) 3470089525 | ✉ orlandoricc@gmail.com | 🏠 riccardorlando.xyz | 📺 riccorl | 🌐 riccardo-orlando | 🎓 Riccardo Orlando

Experience

Researcher

WORLD BANK GROUP

Rome, Italy

Mar. 2025 - Current

- Core member of ImpactAI's modeling team. ImpactAI is an agentic framework that automatically delivers high-quality meta-analyses of development-economics studies to guide policymaking.
- Contributing to the design of pretraining and alignment strategies to adapt Foundation Models to real-world economic impact analysis.

Researcher

SAPIENZA UNIVERSITY OF ROME

Rome, Italy

Nov. 2021 - Jan. 2025

- Led model pretraining and data preprocessing for Minerva (1B, 3B, and 7B) LLMs, the first family of Open Foundation Models trained from scratch on Italian data.
- Developed ReLiK, a novel retrieval-augmented system achieving SOTA performance in Entity Linking and Relation Extraction.
- Led the data annotation process that resulted in a novel dataset for Semantic Role Labeling.
- Developed production-ready pipelines for several state-of-the-art NLU tasks.
- Served as Teaching Assistant for MSc courses in Natural Language Processing.

NLP Engineer

BABELSCAPE

Rome, Italy

Feb. 2021 - Nov. 2021

- Designed and implemented end-to-end multilingual NLP pipelines for production deployment.
- Built scalable ML serving framework that drastically reduced model deployment time.
- Authored and presented two papers at EMNLP 2021.

Software Engineer

CAPGEMINI

Rome, Italy

Mar. 2018 - Sept. 2018

- Developed a back-end application using Spring Boot to handle internal workflows for different customers.
- Contributed to front-end development using Angular.

Education

Ph.D. in Computer Science (Deep Learning & Natural Language Processing)

SAPIENZA UNIVERSITY OF ROME

Rome, Italy

Nov. 2021 - Jan. 2025

- Thesis: Enhancing Semantic Understanding Across Multiple Dimensions: Towards A Unified Framework for Semantic Knowledge Extraction.
- Advisor: Roberto Navigli

Master of Science in Engineering in Computer Science

SAPIENZA UNIVERSITY OF ROME

Rome, Italy

Oct. 2018 - Jan. 2021

- Final Grade: 110/110 with honors
- Thesis: Automatic Approaches to Produce Multilingual Training Data for Semantic Role Labelling.

Bachelor of Science in Computer Science

SAPIENZA UNIVERSITY OF ROME

Rome, Italy

Oct. 2014 - May 2018

- Final Grade: 101/110
- Thesis: Densest Subgraph in Fork/Join.

Relevant Publications

2025	Karim Ghonim, Pere-Lluís Huguet Cabot, <u>Riccardo Orlando</u> , and Roberto Navigli. RAED: Retrieval-Augmented Entity Description Generation for Emerging Entity Linking and Disambiguation.	EMNLP 2025
2024	Francesco Maria Molfese, Simone Conia, <u>Riccardo Orlando</u> , and Roberto Navigli. ZEBRA: Zero-Shot Example-Based Retrieval Augmentation for Commonsense Question Answering.	EMNLP 2024
2024	<u>Riccardo Orlando</u> , Luca Moroni, Pere-Lluís Huguet Cabot, Edoardo Barba, Simone Conia, Sergio Orlandini, Giuseppe Fiameni, and Roberto Navigli. Minerva LLMs: The First Family of Large Language Models Trained from Scratch on Italian Data.	CLiC-it 2024
2024	<u>Riccardo Orlando</u> , Pere-Lluís Huguet Cabot, Edoardo Barba, and Roberto Navigli. ReLiK: Retrieve and Link, Fast and Accurate Entity Linking and Relation Extraction on an Academic Budget.	ACL 2024

2024	Simone Conia, Edoardo Barba, Abelardo Carlos Martínez Lorenzo, Pere-Lluis Huguet Cabot, <u>Ricardo Orlando</u> , Luigi Procopio, Roberto Navigli. MOSAICo: a Multilingual Open-text Semantically Annotated Interlinked Corpus.	NAACL 2024
2023	<u>Riccardo Orlando</u> , Simone Conia, and Roberto Navigli. Exploring Non-Verbal Predicates in Semantic Role Labeling: Challenges and Opportunities.	ACL 2023
2022	<u>Riccardo Orlando</u> , Simone Conia, Stefano Faralli, and Roberto Navigli. Universal Semantic Annotator: the First Unified API for WSD, SRL and Semantic Parsing.	LREC 2022
2021	<u>Riccardo Orlando</u> , Simone Conia, Fabrizio Brignone, Francesco Cecconi, and Roberto Navigli. AMuSE-WSD: An All-in-one Multilingual System for Easy Word Sense Disambiguation.	EMNLP 2021
2021	Simone Conia, <u>Riccardo Orlando</u> , Fabrizio Brignone, Francesco Cecconi, and Roberto Navigli. InVeRo-XL: Making Cross-Lingual Semantic Role Labeling Accessible with Intelligible Verbs and Roles.	EMNLP 2021

Skills

Frameworks	Deep Learning & Machine Learning (PyTorch, PyTorch Lightning, Hydra, Transformers, ONNX, TensorFlow, scikit-learn)
Back-end	FastAPI, Django, REST API
Languages	Python, Java, Kotlin, Bash/Shell
Developer Tools	Git, Docker, VS Code, Visual Studio, PyCharm, IntelliJ

Honors & Awards

2024	Winner , 3x Italian SuperComputing Resource Allocation (ISCRA) Call for Proposal type C for compute hours at CINECA, the Italian HPC Center	Rome, Italy
2023	Winner , Italian SuperComputing Resource Allocation (ISCRA) Call for Proposal type B for compute hours at CINECA, the Italian HPC Center	Rome, Italy
2022	Winner , Call for Proposals for Scientific Research	Rome, Italy

Program Committees

2025	Reviewer , ARR May 2025
2024	Reviewer , ARR February 2024, ARR June 2024
2023	Reviewer , ARR December 2023, ARR October 2023, EMNLP 2023, ACL 2023, EACL 2023
2022	Reviewer , LREC 2022
2021	Reviewer , EMNLP 2021